

ESAM 420 Notes

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1 Motivating Examples

Suppose we want to launch a projectile vertically into the air and we want to create a model. We are launching it vertically so we do not have to worry about vectors. We start with Newton's Law of Gravity so we can say that

$$\frac{d^2r}{dt^2} = -G \frac{M_e}{r^2}$$

where r is the distance from the center of the Earth, G is the gravitational constant, and M_e is the mass of the Earth. However, we want to know how far from the surface of the Earth our projectile goes so we define $r = h + R$ where R is the radius of the Earth and h is the height of the projectile relative to the ground. Since $dr = dh$, we have that

$$\frac{d^2h}{dt^2} = -G \frac{M_e}{(h + R)^2}$$

Recalling that $g = \frac{GM_e}{R^2}$, we rewrite the equation as

$$\frac{d^2h}{dt^2} = -\frac{gR^2}{(h + R)^2}$$

This equation has no analytical solution. However, if make the approximation $h \ll R$, we can obtain an *asymptotic solution*.

$$\frac{d^2h}{dt^2} = -g$$

which if we set the initial conditions to be $h(0) = 0, h'(0) = v_0$, we have that the solution is

$$h(t) = v_0 t - \frac{1}{2} g t^2$$

The problem with this is we have no semi-“precise” way to justify when we can make this assumption. We also cannot add any correcting factors. We usually start these by non-dimensionalizing. This allows us to zoom to the scale we want to work with and justify what is “small.” For the gravity problem, let us define the characteristic length and times to be

$$L_c = \frac{v_0^2}{g} \quad T_c = \frac{v_0}{g}$$

We also define some new normalized variables relative to these characteristic norms (I don't want to use variables and they're mainly about magnitudes, hence norms)

$$x = \frac{h}{L_c} \quad \tau = \frac{t}{T_c}$$

We can transform $\frac{d^2 h}{dt^2} \rightarrow \frac{d^2 x}{d\tau^2}$ via the chain rule

$$\begin{aligned}\frac{dx}{d\tau} &= \frac{dx}{dh} \frac{dh}{dt} \frac{dt}{d\tau} = \frac{1}{L_c} \frac{dh}{dt} \frac{T_c}{1} \\ \frac{dh}{dt} &= \frac{L_c}{T_c} \frac{dx}{d\tau} \\ \Rightarrow \frac{d^2 h}{dt^2} &= \frac{d}{dt} \left(\frac{L_c}{T_c} \frac{dx}{d\tau} \right) = \frac{d}{dt} \frac{dt}{d\tau} \left(\frac{L_c}{T_c} \frac{dx}{d\tau} \right) = \frac{L_c}{T_c^2} \frac{d^2 x}{d\tau^2}\end{aligned}$$

So our equation now becomes

$$\begin{aligned}\frac{L_c}{T_c^2} \frac{d^2 x}{d\tau^2} &= -\frac{gR^2}{(xL_c + R)^2} = -\frac{gR^2}{R^2(x\frac{L_c}{R} + 1)^2} \\ \frac{d^2 x}{d\tau^2} &= -\frac{1}{(x\varepsilon + 1)^2} \quad \varepsilon = \frac{L_c}{R} = \frac{v_0^2}{gR}\end{aligned}$$

We then guess our solution is some sort of expansion based off this small parameter of the form

$$x(t) = x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots$$

Our small parameter being the ratio of the characteristic length to the radius of the Earth shows us exactly when we can make certain assumptions.

Another application is the Navier-Stokes equations.

$$\begin{aligned}\nabla \cdot \vec{u} &= 0 \text{ (incompressibility equation)} \\ \rho \left(\frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla) \vec{u} \right) &= -\nabla p + \mu \nabla^2 u \text{ (equation of motion)}\end{aligned}$$

If we non-sensationalize such that $(\tilde{x}, \tilde{y}, \tilde{z}) = \frac{1}{L}(x, y, z)$, $\tilde{u} = \frac{\vec{u}}{u_c}$, $\tilde{p} = \frac{p}{p_c}$, $\tilde{t} = \frac{t}{T_c}$, $T_c = \frac{L_c}{u_c}$. This transforms our equation into

$$\frac{\rho u_c}{T_c} \frac{\partial \tilde{u}}{\partial \tilde{t}} + \frac{\rho u_c^2}{L_c} (\tilde{u} \cdot \tilde{\nabla}) \tilde{u} = -\frac{p_c}{L_c} \tilde{\nabla} \tilde{p} + \frac{\mu U_c}{L_c^2} \tilde{\nabla}^2 \tilde{u}$$

Notice that the first two coefficients of the left hand side are equal and so we can rewrite the left hand side as one differential operator $\frac{D\tilde{u}}{D\tilde{t}}$. Doing some algebra we get that

$$\text{Re} \frac{D\tilde{u}}{D\tilde{t}} = -\frac{p_c L_c}{\mu u_c} \tilde{\nabla} \tilde{p} + \tilde{\nabla}^2 \tilde{u}$$

where Re is the Reynold's number which is the ratio of inertial versus viscous forces in a flow.

$$\text{Re} = \frac{\rho u_c L_c}{\mu}$$

If we take the limit as the Reynold's number go to zero, we go into the length scale of bacteria typically. If we have $\frac{p_c L_c}{\mu u_c} = 1$, we then have that the characteristic pressure is $p_c = \frac{\mu u_c}{L_c}$. Rearranging the equation we get

$$\nabla p = \mu \nabla^2 \vec{u}$$

which is known as the Stoke's limit. If instead we take the limit as the Reynold's number goes to infinity and if we set the characteristic pressure to be $p_c = \rho u_c^2$, we get

$$\frac{D\vec{u}}{D\tilde{t}} = -\nabla p + \frac{1}{\text{Re}} \tilde{\nabla}^2 \tilde{u}$$

The Laplacian term goes to zero and we have the Euler Equation for inviscid flow

$$\rho \left(\frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla) \vec{u} \right) = -\nabla p$$

From this we can get d'Alembert's paradox which states that the force of drag is zero on a body moving at a constant velocity in an inviscid fluid contrary to observations. Due to asymptotics, we can add correction terms to resolve this paradox.

2 Algebraic Equations

Suppose we want to solve the algebraic equation

$$x^2 + 2\varepsilon x - 1 = 0 \quad \varepsilon \ll 1$$

Analytically we can solve this to get that $x_{1,2} = -\varepsilon \pm \sqrt{1 + \varepsilon^2}$. If we can't analytically solve, we can use two methods to approximate: expansion or iteration. Let us start with the expansion method first.

2.1 Expansion Method

Let us start with the zeroth-order approximation and set $\varepsilon = 0$. We get that our solutions are $x_{0,2} = \pm 1$. We then make the guess that our solution can be expressed as the power series

$$x = x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots$$

We then plug that back into our equation and match together powers of ε up until the powers of epsilon that we desire. Doing so gets us

$$\begin{aligned} (x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots)^2 + 2\varepsilon(x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots) - 1 &= 0 \\ 1 + 2\varepsilon x_1 + 2\varepsilon^2 x_2 + \varepsilon^2 x_1 + \dots + 2\varepsilon + 2\varepsilon^2 x_1 + \dots - 1 &= 0 \end{aligned}$$

Grouping them together by powers of epsilon gives

$$\varepsilon(2x_1 + 2) + \varepsilon^2(2x_2 + x_1^2 + 2x_1) = 0 \quad \forall \varepsilon$$

Since this equation must be true for all ε , we have that the parenthesis terms must be 0 and so we solve for x_1 and then plug that into the ε^2 expression to solve for x_2 .

$$\begin{aligned} \varepsilon : 2x_1 + 2 &= 0 \rightarrow x_1 = -1 \\ \varepsilon^2 : 2x_2 + 1 - 2 &= 0 \rightarrow x_2 = \frac{1}{2} \end{aligned}$$

Our three-term expansion is therefore

$$x = 1 - \varepsilon + \frac{1}{2}\varepsilon^2$$

If we take the binomial expansion of our analytical solution, we find the same thing.

2.2 Iteration Method

We generally solve for the highest order of x

$$x^2 = 1 - 2\varepsilon x \rightarrow x = \pm \sqrt{1 - 2\varepsilon x}$$

The iteration method says that

$$x_{n+1} = \pm \sqrt{1 - 2\varepsilon x_n}$$

We start with the zeroth-order approximation

$$\begin{aligned} \varepsilon = 0 &\rightarrow x_0 = \pm 1 \\ \varepsilon^1 \quad x_1 &= \sqrt{1 - 2\varepsilon} = 1 - \varepsilon \\ \varepsilon^2 \quad x_2 &= \sqrt{1 - 2(1 - \varepsilon)\varepsilon} = 1 - \varepsilon(1 - \varepsilon) - \frac{1}{8}(2\varepsilon(1 - \varepsilon))^2 \end{aligned}$$

Cleaning all of this up we get the same thing as the previous methods

$$x_3 = 1 - \varepsilon + \frac{1}{2}\varepsilon^2$$

The iteration method effectively is a spiral from an initial approximation leading to a value of x_n which we then iterate on more to hopefully get to the desired root. However, if the function changes too much too quickly, this can lead to spiraling away.

The expansion and iteration methods are good when the zeroth-order approximation doesn't change the nature of the equation (e.g., if we have $\varepsilon x^2 + x + 1 = 0$, the zeroth-order approximation would change the equation from a quadratic to a linear equation).

Let us now consider a case when we can't use the highest order term to begin our iteration.

$$x^2 + x - \varepsilon = 0$$

The key thing here is to setup and iteration such that it converges to the desired value. In order for that to happen, we need on an interval I around the root to have the following condition: $|f'(x)| < 1 \ \forall x \in I$. It is also worth noting that knowing what the unperturbed function looks like by moving all of the ε terms to one side and ignoring it (in this case, it would be $x^2 + x = 0$). Here we can see the unperturbed roots are $x = -1$ and $x = 0$. Let us try to find the root around $x = -1$. We can do this with the setup

$$x^2 = \varepsilon - x$$

By dividing by x , we can find the relation

$$x = \frac{\varepsilon}{x_n} - 1$$

If we let the right hand side equal $f(x)$, we find that the derivative is $|f'(x)| = \frac{\varepsilon}{x_n^2}$. Since ε is assumed small, our convergence condition is satisfied.

2.3 Method of Dominant Balance

Let us take a look at the algebraic equation

$$\varepsilon x^2 + x - 1 = 0$$

If we do either the expansion or iteration method, we change the number of roots we get. Let us go try out this method to see what happens. We once again start with the $\varepsilon = 0$ term to get that $a_0 = 1$. We shall use expansion here and guess $x = 1 + \varepsilon a_1 + \varepsilon^2 a_2 + \dots$. Plugging this in we get

$$\begin{aligned} \varepsilon(1 + \varepsilon a_1 + \dots)^2 + (1 + \varepsilon a_1 + \dots) - 1 &= 0 \\ \varepsilon(1 + 2\varepsilon a_1 + \dots) + (1 + \varepsilon a_1 + \dots) - 1 &= 0 \\ \varepsilon(1 + a_1) + \mathcal{O}(\varepsilon^2) &= 0 \rightarrow a_1 = -1 \end{aligned}$$

Therefore our two term expansion is $x = 1 - \varepsilon + \mathcal{O}(\varepsilon^2)$. If we look at the actual roots using the quadratic formula and binomial expand the square root, we get the same thing for the positive root.

$$\begin{aligned} x = \frac{-1 \pm \sqrt{1 + 4\varepsilon}}{2\varepsilon} &= \frac{1}{2\varepsilon} \left(-1 \pm \left(1 + \frac{1}{2}4\varepsilon - \frac{1}{8}(4\varepsilon)^2 + \dots \right) \right) \\ x_+ &= \frac{1}{2\varepsilon}(2\varepsilon - 2\varepsilon^2) = 1 - \varepsilon + \dots \end{aligned}$$

If we look at the negative root, we find that there is a $\frac{1}{\varepsilon}$ term which explains the behavior at $\varepsilon = 0$ and why it doesn't show up.

$$x_- = \frac{1}{2\varepsilon}(-2 - 2\varepsilon + 2\varepsilon^2) = -\frac{1}{\varepsilon} - 1 + \varepsilon + \dots$$

equipped with this knowledge, let us introduce the Method of Dominant Balance. This method allows us to rigorously find the other root by introducing a rescaled variable $y = \frac{x}{\delta}$ where δ is some unknown which may be a function. This relation implies that $x = \delta y$. The reason we are doing this is because in the original equation we are strictly assuming $x = \text{ord}(1)$ with respect to ε and so as such, the linear and constant terms are both order 1 whereas the quadratic term is very small. As such, the dominant balance was between x and -1 and that's why we only recovered one of the roots. This rescaling shifts the dominant balance. Since we know before hand that the irregular root $x \sim \frac{1}{\varepsilon}$, let us plug into the original equation and see what happens.

$$\begin{aligned} \varepsilon \left(\frac{1}{\varepsilon} \right)^2 + \left(\frac{1}{\varepsilon} \right) - 1 &= 0 \\ \frac{1}{\varepsilon} + \frac{1}{\varepsilon} - 1 &= 0 \end{aligned}$$

We can now see that as $\varepsilon \rightarrow 0$, the quadratic and linear terms are now big compared to the constant. That is effectively what the Method of Dominant Balance is doing except we do not have prior knowledge of what this relation is. As such, we substitute our relation for x to arrive at

$$\varepsilon\delta^2y^2 + \delta y - 1 = 0$$

and now we balance the two other terms (with the assumption that y is strictly order 1). For this to happen, we must have the relation

$$\varepsilon\delta^2 \sim \delta$$

which we can then extract the relation $\delta \sim \frac{1}{\varepsilon}$. This is exactly the term that caused the negative root to vanish when we used the quadratic formula to solve for it. We now have the relation $x = \frac{1}{\varepsilon}y$. We are now able to substitute this back in for every x in the original algebraic equation and derive a new one that we can apply our previous methods on.

$$\begin{aligned}\frac{1}{\varepsilon}y^2 + \frac{1}{\varepsilon}y - 1 &= 0 \\ \Rightarrow y^2 + y - \varepsilon &= 0\end{aligned}$$

We cannot find the solutions to this using iteration because we would have the relation $y = \pm\sqrt{\varepsilon - y}$ which if we tried to find the leading term expression, we would result in complex values. That is why we shall use expansion

$$\begin{aligned}y &= a_0 + \varepsilon a_1 + \varepsilon^2 a_2 + \dots \\ (a_0 + \varepsilon a_1 + \dots)^2 + (a_0 + \varepsilon a_1 + \dots) - \varepsilon &= 0 \\ a_0^2 + 2\varepsilon a_0 a_1 + a_0 + \varepsilon a_1 - \varepsilon &= 0 \\ a_0(a_0 + 1) + \varepsilon(2a_0 a_1 + a_1 - 1) &= 0\end{aligned}$$

Solving for a_1 in the ε term we see that

$$a_1 = \frac{1}{2a_0 + 1}$$

We see that $a_0 = 0$ or $a_0 = -1$. In the first case, we have that $a_1 = 1$ and so the expansion for that is $y = \varepsilon + \dots$ and for the second, we have $-1 - \varepsilon + \dots$. Subbing this back into the relation for $x = \frac{1}{\varepsilon}y$, we get the approximations

$$\begin{aligned}x_1 &= 1 + \dots \\ x_2 &= -\frac{1}{\varepsilon} - 1 + \dots\end{aligned}$$

Let us look at another example

$$\varepsilon x^3 - x + 1 = 0$$

Finding the regular root is pretty trivial so we shall ignore it. We start with the ansatz that $x = \delta y$. Subbing this in we get

$$\varepsilon\delta^3y^3 - \delta y + 1 = 0$$

We can choose to either balance the first and second term or the first and third term. If we choose to balance the first and last term, we get that $\delta \sim \varepsilon^{-1/3}$. Plugging this in we would get

$$y^3 - \varepsilon^{-1/3}y + 1 = 0$$

The first and third terms are now both order 1 but the middle term is huge as $\varepsilon \rightarrow 0$ which is impossible since you now have an unbalanced term. Therefore the correct terms to balance were the first and second. We then get $\delta \sim \frac{1}{\sqrt{\varepsilon}}$ and so

$$\frac{1}{\sqrt{\varepsilon}}y^3 - \frac{1}{\sqrt{\varepsilon}}y + 1 = 0 \rightarrow y^3 - y + \sqrt{\varepsilon} = 0.$$

We can now iterate/expand to find the roots we couldn't find earlier. Additionally, it is worth noting since the non-homogeneity is $\sqrt{\varepsilon}$, we should have our expansion be

$$y = y + 0 + \sqrt{\varepsilon}y_1 + (\sqrt{\varepsilon})^2y_2 + \dots$$